

SCADS Software Development Plan

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1 Project Overview

SCADS, Spacecraft Cube Attitude Determination System, is attitude determination flight software for a simple rigid body cube spacecraft in orbit. The end goal is to complete the following objectives:

- Write the real time operating system based on FreeRTOS
- Prove the tasks and ISRs are possible via Rate Monotonic Scheduling Analysis
- Create a simulation environment that can support SIL and HIL simulations, with a 3D visualization component

2 Milestones

The project began April 19, 2026 and will be considered complete when all of the requirements are complete and the repository is in a professional presentable state I am satisfied with.

3 Branching Strategy

Branching will be done with rebase and `-no-ff`, meaning every time a development branch is merged to main, a merge commit will be created, which is required to indicate a point in time where the branch was integrated. Branches should be grouped by overall feature, with a description. For instance:

- docs/short-description : documentation work
- dev/short-description : core code and unit test (CUT) work
- sim/short-description : simulation work
- visual/short-description : 3D visualization work

4 Issues and Defects

Issues will be tracked with Github Issues and worked accordingly. Versions will be managed via git tags.